

Package ‘dynTools’

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BugReports <https://github.com/jeksterslab/dynTools/issues>

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CheckDynData

Check Dynamic Modeling Data

Description

The function checks whether a data frame has the variables and structure required by the dynamic modeling utility functions.

Usage

```

CheckDynData(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  require_unique = TRUE,
  require_numeric_time = TRUE,
  require_numeric_observed = TRUE,
  require_numeric_covariates = FALSE,
  min_rows = 2L
)

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>require_unique</code>	Logical. If TRUE, require unique id-time combinations.
<code>require_numeric_time</code>	Logical. If TRUE, require the time variable to be numeric.
<code>require_numeric_observed</code>	Logical. If TRUE, require observed variables to be numeric.
<code>require_numeric_covariates</code>	Logical. If TRUE, require covariates to be numeric.
<code>min_rows</code>	Positive integer. Minimum number of rows required per ID.

Value

Returns TRUE invisibly if all checks pass.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
CheckDynData(
```

```

data = data,
id = "id",
time = "time",
observed = c("y1", "y2")
)

```

CombineByIDTime

Combine Data Sets by ID and Time

Description

The function appends observed variables from one data set to another data set using exact id-time matches. Rows in data define the output id-time structure. If an id-time row in data is not present in append_data, missing values are inserted for the appended observed variables.

Usage

```
CombineByIDTime(data, append_data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
append_data	Data frame. A data frame object containing the id and time columns and the observed variables to append to data.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  cov = c(10, 10, 10, 20, 20, 20)
)

append_data <- data.frame(
  id = c(1, 1, 2),
  time = c(1, 3, 2),
  y1 = c(11, 13, 22),
  y2 = c(101, 103, 202)
)

CombineByIDTime(
  data = data,
  append_data = append_data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  covariates = "cov"
)
```

DeleteInitialNA

*Delete for NAs in Initial Row By ID***Description**

The function removes initial rows by ID if they contain missing values. This process is repeated until the first row per ID no longer has missing observations.

Usage

```
DeleteInitialNA(data, id, time, observed, covariates = NULL)
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, 10, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, 10, 11, 12, 13, 14),
  y3 = c(1, NA, 3, 4, 5, 10, 11, 12, 13, 14)
)
data

DeleteInitialNA(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3)
)
```

DeleteObservedAllNA *Delete Rows With All Observed Variables Missing*

Description

The function removes rows where all observed variables are missing.

Usage

```
DeleteObservedAllNA(data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Details

Covariates are retained in the returned data, but they are not used to decide whether a row should be deleted.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, NA, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, NA, 11, 12, 13, 14),
  y3 = c(NA, NA, 3, 4, 5, NA, 11, 12, 13, 14)
)
data

DeleteObservedAllNA(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3)
)
```

DeltaTByID

*Summarize Time Intervals by ID***Description**

The function returns a diagnostic table describing the spacing of the time variable within each ID.

Usage

```
DeltaTByID(
  data,
  id,
  time,
  observed = NULL,
  covariates = NULL,
  tolerance = sqrt(.Machine$double.eps)
)
```

Arguments

data	Data frame. A data frame object.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. Optional vector of character strings of the names of the observed variables in the data.
covariates	Character vector. Optional vector of character strings of the names of the covariates in the data.
tolerance	Numeric. Tolerance used to determine whether the time intervals are regular.

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = c(1, 2, 3, 1, 3, 4),
  y = rnorm(6)
)
data

DeltaTByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y"
)
```

DetrendByID

Detrend Observed Variables by ID

Description

The function removes polynomial time trends from observed variables within each ID. Missing observed values are ignored when estimating the trend and remain missing in the detrended variables.

Usage

```
DetrendByID(
  data,
  id,
  time,
  observed,
```

```

    covariates = NULL,
    degree = 1L,
    replace = FALSE,
    keep_mean = TRUE,
    prefix = "detrend",
    warn_skipped = TRUE,
    drop_skipped_ids = TRUE
  )

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>degree</code>	Non-negative integer. Degree of the polynomial trend. Use <code>degree = 0</code> for mean centering by ID when <code>keep_mean = FALSE</code> . If <code>degree = 0</code> and <code>keep_mean = TRUE</code> , the original observed values are returned.
<code>replace</code>	Logical. If TRUE, replace observed variables with detrended variables. If FALSE, append detrended variables to the data frame.
<code>keep_mean</code>	Logical. If TRUE, preserve the original within-ID mean after detrending. If FALSE, return ordinary residuals from the within-ID trend model.
<code>prefix</code>	Character string. Prefix for detrended variables when <code>replace = FALSE</code> .
<code>warn_skipped</code>	Logical. If TRUE, warn when one or more ID-variable combinations cannot be detrended because of too few usable observations, too few unique time values, or a rank-deficient trend design.
<code>drop_skipped_ids</code>	Logical. If TRUE, remove all rows for IDs with at least one observed variable that cannot be detrended.

Details

If `keep_mean = TRUE`, the within-ID observed mean is added back after removing the fitted time trend. This removes temporal drift while preserving each ID's observed-variable level. If `keep_mean = FALSE`, the output is the ordinary residual from the within-ID polynomial trend model.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 4),
  time = rep(1:4, times = 2),
  y = rep(1:4, times = 2)
)
data
```

```
DetrendByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  degree = 1,
  keep_mean = TRUE
)
```

```
DetrendByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  degree = 1,
  keep_mean = FALSE
)
```

Description

The function computes ID-level diagnostics for intensive longitudinal data. Diagnostics include the number of observed rows, number of complete rows, proportion of all-missing observed rows, duplicate ID-time rows, time gaps, within-ID standard deviations, and counts of extreme observed values.

Usage

```

DiagnosticsByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  missing_codes = NULL,
  min_nonmissing = 1L,
  extreme_cut = c(4, 5, 6),
  sd_cut = c(0.1, 0.05),
  time_scale = 1,
  posix_unit = "hours"
)

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>missing_codes</code>	NULL or vector. Values to temporarily treat as missing when computing diagnostics. The input data are not modified.
<code>min_nonmissing</code>	Positive integer. Minimum number of non-missing observed variables required for a row to count as an observed row.
<code>extreme_cut</code>	Numeric vector. Absolute-value cutoffs used to count extreme values.
<code>sd_cut</code>	Numeric vector. Within-ID standard-deviation cutoffs used to count low-variance variables.
<code>time_scale</code>	Positive number. Multiplier applied to numeric time gaps. For example, use <code>time_scale = 24</code> when time is measured in days and gaps should be reported in hours.
<code>posix_unit</code>	Character string. Unit for gaps when time is a POSIXt or Date variable. Passed to <code>as.numeric.difftime()</code> .

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

DiagnosticsByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

DropByID

*Drop IDs From a Data Frame***Description**

The function removes rows belonging to one or more IDs. It is a small convenience wrapper used after ID-level diagnostics identify IDs for exclusion or sensitivity analysis.

Usage

```
DropByID(data, id, drop = NULL)
```

Arguments

data	Data frame.
id	Character string. Name of the ID variable.
drop	Vector. ID values to remove. If NULL or empty, data is returned unchanged.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2, 3, 3),
  y = 1:6
)

DropByID(
  data = data,
  id = "id",
  drop = c(2, 3)
)
```

ElapsedTimeByID	<i>Create Elapsed Time by ID</i>
-----------------	----------------------------------

Description

The function creates an elapsed-time variable within ID. For date-time variables, elapsed time is computed with `difftime()`. For numeric time variables, elapsed time is computed by subtraction, optionally converting from `input_units` to `units`.

Usage

```
ElapsedTimeByID(
  data,
  id,
  time,
  output = "time_elapsed",
  units = "days",
  origin = c("by_id", "global"),
```

```

    input_units = NULL,
    replace = FALSE
  )

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>output</code>	Character string. Name of the elapsed-time variable.
<code>units</code>	Character string. Units for the output elapsed time.
<code>origin</code>	Character string. If <code>origin = "by_id"</code> , each ID's origin is its own minimum non-missing time. If <code>origin = "global"</code> , all IDs use the global minimum non-missing time.
<code>input_units</code>	Character string or NULL. Units of numeric input time. If NULL, numeric time is assumed to already be in the desired output scale and only subtraction is performed.
<code>replace</code>	Logical. If TRUE, replace the original time variable. If FALSE, append output.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```

data <- data.frame(
  id = c(1, 1, 2, 2),
  datetime = as.POSIXct(
    c(
      "2020-01-01 00:00:00",
      "2020-01-02 00:00:00",

```

```

      "2020-01-03 12:00:00",
      "2020-01-04 00:00:00"
    ),
    tz = "UTC"
  )
)
data

ElapsedTimeByID(
  data = data,
  id = "id",
  time = "datetime",
  output = "time",
  units = "days"
)

```

FilterByID

*Filter Dynamic Modeling Data by ID***Description**

The function removes IDs that do not satisfy minimum data requirements.

Usage

```

FilterByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_rows = 2L,
  min_complete = 2L,
  max_prop_missing = 1,
  allow_initial_na = TRUE,
  allow_all_missing = FALSE
)

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.

observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
min_rows	Non-negative integer. Minimum number of rows required per ID.
min_complete	Non-negative integer. Minimum number of complete observed rows required per ID.
max_prop_missing	Numeric. Maximum allowed proportion of missing values across observed variables.
allow_initial_na	Logical. If FALSE, remove IDs where the initial row contains missing values.
allow_all_missing	Logical. If FALSE, remove IDs where all observed values are missing.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:3, each = 3),
  time = rep(1:3, times = 3),
  y = c(1, 2, 3, NA, NA, 4, NA, NA, NA)
)
data

FilterByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  min_complete = 2
)
```

FilterObservedRows *Remove Rows With Too Few Observed Values*

Description

The function removes rows with fewer than a requested number of non-missing observed variables. This is useful after regularizing or padding intensive longitudinal data, where many rows may contain no observed measurements but retain an ID and time value.

Usage

```
FilterObservedRows(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_nonmissing = 1L,
  min_rows_by_id = NULL
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
min_nonmissing	Positive integer. Minimum number of non-missing observed variables required to retain a row.
min_rows_by_id	NULL or positive integer. If not NULL, IDs with fewer than min_rows_by_id retained rows are removed after row filtering.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2),
  time = c(1, 2, 3, 1, 2),
  y1 = c(1, NA, 3, NA, 2),
  y2 = c(NA, NA, 4, NA, 3)
)

FilterObservedRows(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

FlagDiagnosticsByID *Flag Potentially Problematic IDs*

Description

The function adds rule-based flags to the output of [DiagnosticsByID\(\)](#). It is intended for sensitivity checks before fitting dynamic models.

Usage

```
FlagDiagnosticsByID(
  x,
  min_observed_rows = 30L,
  min_complete_rows = NULL,
  max_prop_all_missing = 0.95,
  max_gap = NULL,
  max_median_gap = NULL,
  min_sd = 0.05,
  extreme_cut = 6,
  drop_score_cut = 2L
)
```

Arguments

<code>x</code>	Data frame returned by DiagnosticsByID() .
<code>min_observed_rows</code>	Positive integer. Minimum number of observed rows expected per ID.
<code>min_complete_rows</code>	NULL or positive integer. Minimum number of complete rows expected per ID.
<code>max_prop_all_missing</code>	NULL or number between 0 and 1. Maximum tolerated proportion of all-missing observed rows.
<code>max_gap</code>	NULL or positive number. Maximum tolerated observed-row time gap in the units used by DiagnosticsByID() .
<code>max_median_gap</code>	NULL or positive number. Maximum tolerated median observed-row time gap in the units used by DiagnosticsByID() .
<code>min_sd</code>	Non-negative number. Minimum tolerated within-ID standard deviation across observed variables.
<code>extreme_cut</code>	Positive number. Absolute-value cutoff used to flag extreme observations.
<code>drop_score_cut</code>	Positive integer. IDs with priority scores greater than or equal to this value are marked as sensitivity-drop candidates.

Value

Returns `x` with flag columns appended.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

diagnostics <- DiagnosticsByID(
  data = data,
```

```

    id = "id",
    time = "time",
    observed = c("y1", "y2")
  )

  FlagDiagnosticsByID(
    x = diagnostics,
    min_observed_rows = 3,
    min_sd = 0.05
  )

```

GetDropID

*Get Sensitivity-Drop IDs From Diagnostics***Description**

The function extracts IDs marked as sensitivity-drop candidates by [FlagDiagnosticsByID\(\)](#).

Usage

```
GetDropID(x, flagged_only = TRUE)
```

Arguments

<code>x</code>	Data frame returned by FlagDiagnosticsByID() .
<code>flagged_only</code>	Logical. If TRUE, return IDs marked as sensitivity-drop candidates. If FALSE, return all flagged IDs.

Value

Returns a vector of IDs.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```

data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)

diagnostics <- DiagnosticsByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)

flagged <- FlagDiagnosticsByID(
  x = diagnostics,
  min_observed_rows = 3,
  min_sd = 0.05
)

GetDropID(flagged)

```

GetObservedInitial	<i>Get Observed Initial Means and Covariances</i>
--------------------	---

Description

The function extracts the first observed time point for each ID and computes the sample mean vector and sample covariance matrix of the observed variables across IDs. The resulting mean vector and covariance matrix can be used as fixed initial conditions in dynamic models.

Usage

```
GetObservedInitial(data, id, time, observed, ridge = 1e-06)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.

ridge Positive numeric. Small value added to the diagonal of the covariance matrix if the estimated covariance matrix is not positive definite.

Details

Rows are sorted by ID and time before selecting the first row for each ID. Missing observed values are ignored when computing the mean vector. The covariance matrix is computed using complete cases of the selected first-time observations.

If the covariance matrix is not positive definite, a small ridge value is added to the diagonal.

Value

Returns a list with the following elements:

data Data frame containing the first observed time point for each ID and the selected observed variables.

mean Numeric vector of observed-variable means computed from the first observed time point across IDs.

cov Observed-variable covariance matrix computed from complete cases of the first observed time point across IDs.

n Number of IDs with a first observed row.

n_complete Number of IDs with complete data on all selected observed variables at the first observed time point.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(1, 2, 3, 4, 2, 3, 4, 5, 3, 4, 5, 6),
  y2 = c(4, 3, 2, 1, 5, 4, 3, 2, 6, 5, 4, 3)
)
data

init <- GetObservedInitial(
  data = data,
```

```

    id = "id",
    time = "time",
    observed = c("y1", "y2")
  )

  init$data
  init$mean
  init$cov
  init$n
  init$n_complete

```

InitialNA

*Check for NAs in Initial Row By ID***Description**

The function checks if there are missing values for the initial row by ID.

Usage

```
InitialNA(data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a vector of ID numbers where the initial row has any missing value.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 5),
  time = rep(1:5, times = 2),
  y1 = c(NA, NA, 3, 4, 5, 10, 11, 12, 13, 14),
  y2 = c(NA, 2, 3, 4, 5, 10, 11, 12, 13, 14),
  y3 = c(1, NA, 3, 4, 5, 10, 11, 12, 13, 14)
)
data

InitialNA(
  data = data,
  id = "id",
  time = "time",
  observed = paste0("y", 1:3)
)
```

InsertNA	<i>Insert NAs for Missing Observations</i>
----------	--

Description

The function creates a sequence of time values. It starts with the smallest time value as the starting point and the largest time value as the endpoint. The sequence is incremented by delta_t. This new sequence is combined with the existing empirical time values. For any specific time value where there are no observations, NAs are inserted.

Usage

```
InsertNA(data, id, time, observed, covariates = NULL, delta_t)
```

Arguments

- data Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
- id Character string. A character string of the name of the ID variable in the data.

time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
delta_t	Positive number. Time interval.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 4, 1, 3, 4),
  y1 = c(10, 11, 13, 20, 22, 23),
  y2 = c(5, 6, 8, 15, 17, 18)
)
data

InsertNA(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  delta_t = 1
)
```

LagByID

*Create Lagged Variables by ID***Description**

The function creates lagged observed variables within ID. Lagged values never cross ID boundaries.

Usage

```
LagByID(data, id, time, observed, covariates = NULL, lags = 1L, prefix = "lag")
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
lags	Positive integer vector. Lags to create.
prefix	Character string. Prefix for the lagged variable names.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = rnorm(6),
  y2 = rnorm(6)
)
data

LagByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  lags = 1
)
```

MakeClockTime

Make Clock Time

Description

The function combines a date and a clock time into a POSIXct vector. Clock time can be supplied as decimal hours, for example 13.5, or as a character time, for example "13:30" or "13:30:00".

Usage

```
MakeClockTime(
  date,
  time,
  tz = "UTC",
  date_formats = c("%m/%d/%y", "%m/%d/%Y", "%Y-%m-%d"),
  invalid = c("NA", "error")
)
```

Arguments

date	Date vector. Dates can be supplied as Date, POSIXt, or character values.
time	Numeric or character vector. Clock time in decimal hours, "HH:MM", or "HH:MM:SS" format.
tz	Character string. Time zone passed to as.POSIXct().
date_formats	Character vector. Date formats used to parse character dates.
invalid	Character string. If invalid = "NA", invalid clock times are returned as NA. If invalid = "error", invalid clock times throw an error.

Value

Returns a POSIXct vector.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
MakeClockTime(
  date = "2020-01-02",
  time = 13.5,
  tz = "America/New_York"
)
```

```
MakeClockTime(
  date = "01/02/20",
  time = "13:30",
  tz = "America/New_York"
)
```

PlotByID

Plot Observed Variables by ID

Description

The function creates one time-series plot for each observed variable. Within each plot, trajectories are overlaid by ID.

Usage

```
PlotByID(
  data,
  id,
  time,
  observed,
  ids = NULL,
  times = NULL,
```

```

    type = "b",
    pch = NULL,
    lty = NULL,
    col = NULL,
    xlab = NULL,
    ylab = NULL,
    main = NULL,
    xlim = NULL,
    ylim = NULL,
    legend = FALSE,
    ask = NULL,
    ...
  )

```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>ids</code>	Optional vector. ID values to plot. If NULL, all available IDs are plotted.
<code>times</code>	Optional vector of time values. If NULL, all available time points are plotted. If supplied, the range of times is used to subset the data. For example, <code>times = c(0, 9)</code> plots all rows with time values from 0 to 9, inclusive.
<code>type</code>	Character string. Plot type passed to <code>graphics::lines()</code> . Default is "b".
<code>pch</code>	Optional vector of plotting characters. If NULL, plotting characters are generated automatically by ID. Recycled to match the number of plotted IDs.
<code>lty</code>	Optional vector of line types. If NULL, line types are generated automatically by ID. Recycled to match the number of plotted IDs.
<code>col</code>	Optional vector of colors. If NULL, colors are generated automatically by ID. Recycled to match the number of plotted IDs.
<code>xlab</code>	Optional character string. Label for the x-axis. If NULL, the name of the time variable is used.
<code>ylab</code>	Optional character string. Label for the y-axis. If NULL, the observed variable name is used.
<code>main</code>	Optional character string. Plot title prefix. If NULL, the observed variable name is used.
<code>xlim</code>	Optional vector of length 2. Limits for the x-axis.
<code>ylim</code>	Optional numeric vector of length 2. Limits for the y-axis. If NULL, limits are computed separately for each observed variable.

legend	Logical. If TRUE, add a legend identifying IDs.
ask	Logical. If TRUE, ask before drawing the next observed-variable plot. The default is TRUE in interactive sessions when plotting more than one observed variable.
...	Additional arguments passed to <code>graphics::lines()</code> .

Value

Invisibly returns the IDs that were plotted.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PreprocessDynData()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `ScreenByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = rep(1:4, each = 5),
  time = rep(1:5, times = 4),
  y1 = c(
    1, 2, 3, 4, 5,
    2, 3, 4, 5, 6,
    3, 4, 5, 6, 7,
    4, 5, 6, 7, 8
  ),
  y2 = c(
    5, 4, 3, 2, 1,
    6, 5, 4, 3, 2,
    7, 6, 5, 4, 3,
    8, 7, 6, 5, 4
  ),
  y3 = c(
    1, 1, 2, 2, 3,
    2, 2, 3, 3, 4,
    3, 3, 4, 4, 5,
    4, 4, 5, 5, 6
  )
)
data

PlotByID(
  data = data,
```

```

    id = "id",
    time = "time",
    observed = paste0("y", 1:3),
    ask = FALSE
  )

  PlotByID(
    data = data,
    id = "id",
    time = "time",
    observed = paste0("y", 1:3),
    ids = 1:3,
    times = c(1, 3),
    legend = TRUE,
    ask = FALSE
  )

```

PreprocessDynData

Preprocess Intensive Longitudinal Data for Dynamic Modeling

Description

The function implements a reproducible preprocessing workflow for intensive longitudinal data before dynamic model fitting. The workflow can replace missing-value codes, construct a date-time variable, resolve exact duplicate ID-time rows, remove initial all-missing rows, create elapsed time by ID, compute ID-level diagnostics, optionally drop flagged IDs, optionally insert missing rows on a regular grid, detrend observed variables by ID, and center or standardize observed variables by ID.

Usage

```

PreprocessDynData(
  data,
  id,
  observed,
  covariates = NULL,
  date = NULL,
  clock_time = NULL,
  time = NULL,
  tz = "UTC",
  date_formats = c("%m/%d/%y", "%m/%d/%Y", "%Y-%m-%d"),
  invalid = c("NA", "error"),
  datetime_output = "time_obs",
  raw_time_output = "time_raw",
  output_time = "time",
  missing_codes = c(-999, "-999"),
  missing_columns = NULL,

```



```

drop_missing_time = TRUE,
duplicate_method = c("max_complete", "first", "last"),
delete_initial_na = TRUE,
initial_min_nonmissing = 1L,
elapsed_units = "days",
elapsed_origin = c("by_id", "global"),
elapsed_input_units = NULL,
screen = TRUE,
drop_id = NULL,
drop_flagged = FALSE,
min_observed_rows = 30L,
min_complete_rows = NULL,
max_prop_all_missing = 0.95,
max_gap = NULL,
max_median_gap = NULL,
min_sd = 0.05,
flag_extreme_cut = 6,
drop_score_cut = 2L,
min_nonmissing = 1L,
extreme_cut = c(4, 5, 6),
sd_cut = c(0.1, 0.05),
screen_time_scale = 24,
posix_unit = "hours",
regularize = FALSE,
delta_t = NULL,
grid = c("by_id", "global"),
regularize_method = c("preserve", "snap"),
detrend = TRUE,
degree = 1L,
keep_mean = TRUE,
warn_skipped = TRUE,
drop_skipped_ids = TRUE,
center = TRUE,
scale = TRUE,
obs_skip = NULL,
cov_skip = NULL,
keep_raw_time = FALSE,
keep_datetime = TRUE,
final_diagnostics = TRUE,
return_list = TRUE
)

```

Arguments

<code>data</code>	Data frame.
<code>id</code>	Character string. Name of the ID variable.
<code>observed</code>	Character vector. Names of observed variables to preprocess.

covariates	Character vector or NULL. Names of covariates to retain and optionally center or scale.
date	Character string or NULL. Name of the date variable. Use with <code>clock_time</code> to construct clock time.
clock_time	Character string or NULL. Name of the within-day clock-time variable. Use with <code>date</code> .
time	Character string or NULL. Name of an existing time variable. Used when <code>date</code> and <code>clock_time</code> are not supplied.
tz	Character string. Time zone passed to MakeClockTime() .
date_formats	Character vector. Date formats passed to MakeClockTime() .
invalid	Character string. Invalid clock-time handling passed to MakeClockTime() .
datetime_output	Character string. Name of the constructed POSIXct date-time variable when <code>date</code> and <code>clock_time</code> are supplied.
raw_time_output	Character string. Name of the numeric raw-time variable used for duplicate resolution.
output_time	Character string. Name of the elapsed-time variable used for modeling.
missing_codes	NULL or vector. Missing-value codes to replace with NA.
missing_columns	Character vector or NULL. Columns where missing-value codes should be replaced. If NULL, the replacement is applied to observed and covariates.
drop_missing_time	Logical. If TRUE, remove rows with missing constructed/raw time before duplicate resolution.
duplicate_method	Character string. Method passed to ResolveDuplicateIDTime() .
delete_initial_na	Logical. If TRUE, remove leading rows with too few observed values within ID.
initial_min_nonmissing	Positive integer. Minimum number of non-missing observed variables required in the first retained row within ID. Use <code>length(observed)</code> to mimic the stricter behavior of DeleteInitialNA() .
elapsed_units	Character string. Units for elapsed time passed to ElapsedTimeByID() .
elapsed_origin	Character string. Origin passed to ElapsedTimeByID() .
elapsed_input_units	Character string or NULL. Units of numeric raw time passed to ElapsedTimeByID() .
screen	Logical. If TRUE, compute ID-level diagnostics and drop candidates before optional regularization, detrending, and scaling.
drop_id	Vector or NULL. IDs to remove regardless of diagnostic flags.
drop_flagged	Logical. If TRUE, remove IDs selected by ScreenByID() . If FALSE, return the candidates but keep them in the processed data.

min_observed_rows	Positive number. Minimum number of rows with at least min_nonmissing observed variables required for each ID. Passed to ScreenByID() .
min_complete_rows	Positive number or NULL. Minimum number of complete observed rows required for each ID. Passed to ScreenByID() .
max_prop_all_missing	Numeric value in $[0, 1]$ or NULL. Maximum allowed proportion of rows with all observed variables missing for each ID. Passed to ScreenByID() .
max_gap	Positive number or NULL. Maximum allowed observed-time gap for each ID. Passed to ScreenByID() .
max_median_gap	Positive number or NULL. Maximum allowed median observed-time gap for each ID. Passed to ScreenByID() .
min_sd	Non-negative number or NULL. Minimum allowed within-ID standard deviation across observed variables. Passed to ScreenByID() .
flag_extreme_cut	Positive number. Absolute standardized-value cutoff used to flag extreme values. Passed to ScreenByID() .
drop_score_cut	Positive number. Priority-score cutoff used by ScreenByID() when flagged_only = TRUE.
min_nonmissing	Positive integer. Minimum number of non-missing observed variables defining an observed row. Passed to ScreenByID() .
extreme_cut	Numeric vector. Extreme-value cut points used by ScreenByID() .
sd_cut	Numeric vector. Low-variability cut points used by ScreenByID() .
screen_time_scale	Positive number. Multiplier applied to time gaps in ScreenByID() .
posix_unit	Character string. Unit used for POSIXct time-gap calculations in ScreenByID() .
regularize	Logical. If TRUE, call RegularizeTimeByID() . Continuous-time models that handle unequal intervals usually do not require this step.
delta_t	Positive number or NULL. Time interval passed to RegularizeTimeByID() when regularize = TRUE.
grid	Character string. Grid type passed to RegularizeTimeByID() .
regularize_method	Character string. Method passed to RegularizeTimeByID() .
detrend	Logical. If TRUE, call DetrendByID() .
degree	Non-negative integer. Polynomial degree passed to DetrendByID() .
keep_mean	Logical. Passed to DetrendByID() .
warn_skipped	Logical. Passed to DetrendByID() .
drop_skipped_ids	Logical. Passed to DetrendByID() .
center	Logical. If TRUE, call ScaleByID() .
scale	Logical. If TRUE, standardize by ID. If FALSE, only mean-center by ID. Ignored when center = FALSE.

obs_skip, cov_skip
 Character vectors or NULL. Variables to skip when centering or scaling.

keep_raw_time Logical. If FALSE, remove raw_time_output from the returned data.

keep_datetime Logical. If FALSE, remove datetime_output from the returned data.

final_diagnostics
 Logical. If TRUE, compute diagnostics on the final processed data.

return_list Logical. If TRUE, return a list with data, diagnostics, and dropped IDs. If FALSE, return only the processed data.

Value

If return_list = TRUE, returns a list with elements data, diagnostics, final_diagnostics, drop_id, flagged_id, manual_drop_id, and call. If return_list = FALSE, returns the processed data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  date = rep("2020-01-01", 6),
  clock = c("08:00", "09:00", "10:00", "08:00", "09:00", "10:00"),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, 2, 4, 2, 2, 2)
)
```

```
PreprocessDynData(
  data = data,
  id = "id",
  date = "date",
  clock_time = "clock",
  observed = c("y1", "y2"),
  min_observed_rows = 3,
  detrend = FALSE,
  center = FALSE
)
```

RegularizeTimeByID *Regularize Time by ID*

Description

The function inserts rows with missing values so that time follows a regular grid. A global grid uses the same time sequence for every ID. An ID-specific grid uses each ID's own minimum and maximum observed time values. If `method = "preserve"`, the function completes the regular grid while retaining empirical off-grid time values. If `method = "snap"`, empirical time values are snapped to the nearest grid point and the returned data use only grid time values.

Usage

```
RegularizeTimeByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  delta_t,
  grid = c("global", "by_id"),
  method = c("preserve", "snap")
)
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.
<code>delta_t</code>	Positive number. Time interval.
<code>grid</code>	Character string. If <code>grid = "global"</code> , use one time grid from the global minimum to the global maximum time value. If <code>grid = "by_id"</code> , use an ID-specific time grid from each ID's minimum to maximum time value.
<code>method</code>	Character string. If <code>method = "preserve"</code> , complete the regular grid but preserve observed off-grid time values. If <code>method = "snap"</code> , snap observed time values to the nearest grid point so the output contains only regular grid times. When multiple rows for the same ID snap to the same grid point, the row closest

to the grid point is kept; ties are broken by the largest number of non-missing observed/covariate values and then by original order.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 2, 2),
  time = c(1, 3, 1, 2),
  y = c(10, 12, 20, 21)
)
data
```

```
RegularizeTimeByID(
  data = data,
  id = "id",
  time = "time",
  observed = "y",
  delta_t = 1,
  grid = "by_id"
)
```

```
RegularizeTimeByID(
  data = data.frame(
    id = c(1, 1),
    time = c(1.0, 1.7),
    y = c(10, 17)
  ),
  id = "id",
  time = "time",
  observed = "y",
  delta_t = 0.5,
  grid = "by_id",
  method = "snap"
)
```

RegularizeTimeSmallestDeltaTByID

Regularize Time By ID

Description

The function computes the smallest positive time difference across IDs and uses that global delta time to regularize the time grid within each ID. For each ID, a complete sequence from the first to the last observed time is created using the global smallest delta time. Rows not present in the original data are inserted with missing values for all non-ID and non-time variables.

Usage

```
RegularizeTimeSmallestDeltaTByID(
  data,
  id,
  time,
  tol = sqrt(.Machine$double.eps)
)
```

Arguments

<code>data</code>	Data frame.
<code>id</code>	Character string. ID variable.
<code>time</code>	Character string. Time variable.
<code>tol</code>	Numeric. Tolerance used to check whether observed times align with the regularized time grid.

Value

Returns a data frame. The global smallest delta time is stored as the `delta_t` attribute.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(0, 2, 4, 0, 1, 2),
  y = c(1, 2, 3, 10, 11, 12)
)
RegularizeTimeSmallestDeltaTByID(
  data = data,
  id = "id",
  time = "time"
```

)

ReplaceMissingCode *Replace Missing-Value Codes*

Description

The function replaces user-specified missing-value codes with NA in a data frame. The replacement is applied column by column and preserves the original column classes whenever possible.

Usage

```
ReplaceMissingCode(data, values = c(-999, "-999"), columns = names(data))
```

Arguments

data	Data frame.
values	Vector. Values to replace with NA.
columns	Character vector. Names of the columns where replacement should be applied. If NULL, no columns are modified.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = 1:3,
  y = c(1, -999, 3),
  x = c("a", "-999", "c"),
  stringsAsFactors = FALSE
)
data
```



```
ReplaceMissingCode(  
  data = data,  
  values = c(-999, "-999")  
)
```

ResolveDuplicateIDTime

Resolve Duplicate ID-Time Rows

Description

The function resolves exact duplicate id-time rows using a simple deterministic rule. This is useful before inserting missing rows on a time grid because grid-completion functions generally require unique id-time combinations.

Usage

```
ResolveDuplicateIDTime(  
  data,  
  id,  
  time,  
  observed = NULL,  
  covariates = NULL,  
  method = c("first", "last", "max_complete"),  
  order_by = NULL,  
  decreasing = FALSE  
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

method	Character string. Rule used to choose one row from each duplicated id-time group. If method = "first", keep the first row after sorting by id and time. If method = "last", keep the last row after sorting by id and time. If method = "max_complete", keep the row with the largest number of non-missing values in observed.
order_by	Character vector. Optional columns used to break ties after the main method rule.
decreasing	Logical. If TRUE, sort order_by columns in decreasing order.

Value

Returns a data frame with unique id-time rows.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = c(1, 1, 1),
  time = c(0, 0, 1),
  y1 = c(NA, 1, 2),
  y2 = c(NA, 1, 3)
)
data

ResolveDuplicateIDTime(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  method = "max_complete"
)
```

RoundClockTime

*Round Clock Time***Description**

The function rounds, floors, or ceilings clock times to a specified unit. It supports POSIXt, Date, and numeric time values.

Usage

```
RoundClockTime(
  x,
  unit = "hour",
  method = c("round", "floor", "ceiling"),
  origin = "1970-01-01",
  tz = "UTC"
)
```

Arguments

x	A POSIXt, Date, or numeric vector.
unit	Character string. Unit to round to. Supported values are "second", "minute", "hour", "day", and their plural forms.
method	Character string. One of "round", "floor", or "ceiling".
origin	Origin for numeric time values. Only used if x is numeric.
tz	Character string. Time zone used when numeric time values are converted to POSIXct.

Value

Returns a vector with rounded times.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
x <- as.POSIXct(
  "2020-01-01 10:31:00",
  tz = "America/New_York"
)

RoundClockTime(
  x = x,
  unit = "hour"
)
```

ScaleByID

*Scale by ID***Description**

The function scales the data by ID.

Usage

```
ScaleByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  scale = TRUE,
  obs_skip = NULL,
  cov_skip = NULL
)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

scale	Logical. If scale = TRUE, standardize by id. If scale = FALSE, mean center by id.
obs_skip	Character vector. A vector of character strings of the names of the observed variables to skip centering/scaling.
cov_skip	Character vector. A vector of character strings of the names of the covariates to skip centering/scaling.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(
    1, 2, 3, 4,
    10, 11, 12, 13,
    20, 22, 24, 26
  ),
  y2 = c(
    4, 3, 2, 1,
    13, 12, 11, 10,
    26, 24, 22, 20
  )
)
data

ScaleByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)

ScaleByID(
  data = data,
```

```

    id = "id",
    time = "time",
    observed = c("y1", "y2"),
    scale = FALSE
  )

```

ScreenByID

Screen IDs Using Dynamic-Data Diagnostics

Description

The function computes ID-level diagnostics, applies rule-based flags, and returns IDs that are candidates for exclusion or sensitivity analysis. It is a wrapper around [DiagnosticsByID\(\)](#), [FlagDiagnosticsByID\(\)](#), and [GetDropID\(\)](#).

Usage

```

ScreenByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  missing_codes = NULL,
  min_nonmissing = 1L,
  extreme_cut = c(4, 5, 6),
  sd_cut = c(0.1, 0.05),
  time_scale = 1,
  posix_unit = "hours",
  min_observed_rows = 30L,
  min_complete_rows = NULL,
  max_prop_all_missing = 0.95,
  max_gap = NULL,
  max_median_gap = NULL,
  min_sd = 0.05,
  flag_extreme_cut = 6,
  drop_score_cut = 2L,
  flagged_only = TRUE
)

```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
------	---

id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
missing_codes	NULL or vector. Values to temporarily treat as missing when computing diagnostics. The input data are not modified.
min_nonmissing	Positive integer. Minimum number of non-missing observed variables required for a row to count as an observed row.
extreme_cut	Numeric vector. Absolute-value cutoffs used to count extreme values.
sd_cut	Numeric vector. Within-ID standard-deviation cutoffs used to count low-variance variables.
time_scale	Positive number. Multiplier applied to numeric time gaps. For example, use <code>time_scale = 24</code> when time is measured in days and gaps should be reported in hours.
posix_unit	Character string. Unit for gaps when time is a POSIXt or Date variable. Passed to <code>as.numeric.difftime()</code> .
min_observed_rows	Positive integer. Minimum number of observed rows expected per ID.
min_complete_rows	NULL or positive integer. Minimum number of complete rows expected per ID.
max_prop_all_missing	NULL or number between 0 and 1. Maximum tolerated proportion of all-missing observed rows.
max_gap	NULL or positive number. Maximum tolerated observed-row time gap in the units used by DiagnosticsByID() .
max_median_gap	NULL or positive number. Maximum tolerated median observed-row time gap in the units used by DiagnosticsByID() .
min_sd	Non-negative number. Minimum tolerated within-ID standard deviation across observed variables.
flag_extreme_cut	Positive number. Absolute-value cutoff used by FlagDiagnosticsByID() . This value is added to <code>extreme_cut</code> before calling DiagnosticsByID() so that the required total-count column is available.
drop_score_cut	Positive integer. IDs with priority scores greater than or equal to this value are marked as sensitivity-drop candidates.
flagged_only	Logical. If TRUE, return IDs marked as sensitivity-drop candidates. If FALSE, return all flagged IDs.

Value

Returns a list with the following elements:

`diagnostics` Diagnostics with flag columns appended.

`drop_id` IDs selected by `GetDropID()`.

`keep_id` IDs not in `drop_id`.

`call` The matched call.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: `CheckDynData()`, `CombineByIDTime()`, `DeleteInitialNA()`, `DeleteObservedAllNA()`, `DeltaTByID()`, `DetrendByID()`, `DiagnosticsByID()`, `DropByID()`, `ElapsedTimeByID()`, `FilterByID()`, `FilterObservedRows()`, `FlagDiagnosticsByID()`, `GetDropID()`, `GetObservedInitial()`, `InitialNA()`, `InsertNA()`, `LagByID()`, `MakeClockTime()`, `PlotByID()`, `PreprocessDynData()`, `RegularizeTimeByID()`, `ReplaceMissingCode()`, `ResolveDuplicateIDTime()`, `RoundClockTime()`, `ScaleByID()`, `SubsetByID()`, `SummaryByID()`, `TrimInitialRowsByID()`

Examples

```
data <- data.frame(
  id = c(1, 1, 1, 2, 2, 2),
  time = c(1, 2, 3, 1, 2, 3),
  y1 = c(1, NA, 3, 1, 1, 1),
  y2 = c(NA, NA, 4, 2, 2, 2)
)
```

```
ScreenByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  min_observed_rows = 3,
  min_sd = 0.05
)
```

SmallestDeltaTByID

Smallest Global Delta Time By ID

Description

The function computes the smallest positive time difference between consecutive unique time points within ID, then takes the minimum across IDs.

Usage

```
SmallestDeltaTByID(data, id, time)
```

Arguments

data	Data frame.
id	Character string. ID variable.
time	Character string. Time variable.

Value

Returns a numeric value.

Author(s)

Ivan Jacob Agaloos Pesigan

Examples

```
data <- data.frame(  
  id = c(1, 1, 1, 2, 2, 2),  
  time = c(0, 2, 4, 0, 1, 2),  
  y = c(1, 2, 3, 10, 11, 12)  
)  
SmallestDeltaTByID(  
  data = data,  
  id = "id",  
  time = "time"  
)
```

SubsetByID

Subset Data Set by ID

Description

The function creates a list of data frames for each ID.

Usage

```
SubsetByID(data, id, time, observed, covariates = NULL)
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.
<code>observed</code>	Character vector. A vector of character strings of the names of the observed variables in the data.
<code>covariates</code>	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a list by ID numbers.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SummaryByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:3, each = 4),
  time = rep(1:4, times = 3),
  y1 = c(
    1, 2, 3, 4,
    10, 11, 12, 13,
    20, 21, 22, 23
  ),
  y2 = c(
    4, 3, 2, 1,
    13, 12, 11, 10,
    23, 22, 21, 20
  )
)
data

SubsetByID(
```

```

data = data,
id = "id",
time = "time",
observed = c("y1", "y2")
)

```

SummaryByID

Summarize Dynamic Modeling Data by ID

Description

The function returns a diagnostic table with one row per ID.

Usage

```
SummaryByID(data, id, time, observed, covariates = NULL)
```

Arguments

data	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
id	Character string. A character string of the name of the ID variable in the data.
time	Character string. A character string of the name of the TIME variable in the data.
observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.

Value

Returns a data frame with one row per ID.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [TrimInitialRowsByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 3),
  time = rep(1:3, times = 2),
  y1 = c(1, NA, 3, 4, 5, 6),
  y2 = c(1, 2, 3, NA, NA, NA)
)
data

SummaryByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2")
)
```

TrimInitialRowsByID *Trim Leading Rows With Too Few Observed Values by ID*

Description

The function removes leading rows within each ID until the first retained row has at least `min_nonmissing` non-missing observed variables. This is useful before creating an elapsed-time variable when leading records contain no usable measurements. Unlike `DeleteInitialNA()`, the default does not require the first retained row to be complete on all selected variables.

Usage

```
TrimInitialRowsByID(
  data,
  id,
  time,
  observed,
  covariates = NULL,
  min_nonmissing = 1L
)
```

Arguments

<code>data</code>	Data frame. A data frame object of data for potentially multiple subjects that contain a column of subject ID numbers (i.e., an ID variable), a column indicating subject-specific measurement occasions (i.e., a TIME variable), at least one column of observed values.
<code>id</code>	Character string. A character string of the name of the ID variable in the data.
<code>time</code>	Character string. A character string of the name of the TIME variable in the data.

observed	Character vector. A vector of character strings of the names of the observed variables in the data.
covariates	Character vector. A vector of character strings of the names of the covariates in the data.
min_nonmissing	Positive integer. Minimum number of non-missing observed variables required in the first retained row for each ID.

Value

Returns a data frame.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Dynamic Modeling Utility Functions: [CheckDynData\(\)](#), [CombineByIDTime\(\)](#), [DeleteInitialNA\(\)](#), [DeleteObservedAllNA\(\)](#), [DeltaTByID\(\)](#), [DetrendByID\(\)](#), [DiagnosticsByID\(\)](#), [DropByID\(\)](#), [ElapsedTimeByID\(\)](#), [FilterByID\(\)](#), [FilterObservedRows\(\)](#), [FlagDiagnosticsByID\(\)](#), [GetDropID\(\)](#), [GetObservedInitial\(\)](#), [InitialNA\(\)](#), [InsertNA\(\)](#), [LagByID\(\)](#), [MakeClockTime\(\)](#), [PlotByID\(\)](#), [PreprocessDynData\(\)](#), [RegularizeTimeByID\(\)](#), [ReplaceMissingCode\(\)](#), [ResolveDuplicateIDTime\(\)](#), [RoundClockTime\(\)](#), [ScaleByID\(\)](#), [ScreenByID\(\)](#), [SubsetByID\(\)](#), [SummaryByID\(\)](#)

Examples

```
data <- data.frame(
  id = rep(1:2, each = 4),
  time = rep(1:4, times = 2),
  y1 = c(NA, 1, 2, 3, NA, NA, 1, 2),
  y2 = c(NA, NA, 2, 3, NA, 1, 1, 2)
)

TrimInitialRowsByID(
  data = data,
  id = "id",
  time = "time",
  observed = c("y1", "y2"),
  min_nonmissing = 1
)
```

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